

Large Language Models for Trading

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Note: This lecture is majorly based on this [Quantra Course](#)



The Paradigm Shift

Traditional Quantitative Trading

Relied almost entirely on structured numerical data: price, volume, fundamentals.

Text was too difficult to process accurately at scale.

LLM Revolution

Process complex natural language with human-like comprehension at machine speed.

Extract alpha from news, earnings calls, and research reports.

From Predictive to Generative

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1

Traditional ML

Predicting numbers (price > X) or
classifying data points

2

Generative AI

Generating net-new sequences: text,
code, structured data formats

3

Core Mechanism

Next-token prediction: sophisticated
probability engines predicting most
likely next word

Why now? Compute power, massive datasets, and architectural breakthroughs converged.

What is Generative AI?

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1. ML model that can create new content dynamically after being trained
2. Can generate complex types of outputs
3. Inspiration: Machine translation models
4. LLMs are a type of Generative AI

Reads on Gen AI

1. https://www.researchgate.net/publication/379219460_A_Comprehensive_Review_of_Generative_AI_-From_its_Origins_to_Today_and_Beyond
2. <https://arxiv.org/abs/2405.11029>
3. <https://www.sciencedirect.com/science/article/pii/S2667241323000198>

On Machine Translation...

On Machine Translation...

1. Translating text or speech from one language to another
2. Initial approach: using rules and logic
3. Problem: Exceptions in language rules
4. Improvement: Trained on human translations
5. Now: Combining massive amounts of text data with human translations

How do LLMs work?

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1. Text to numerical vector encoding
2. Processing to recognize patterns, and provide context
3. Numerical vector to text decoding

How do LLMs work?

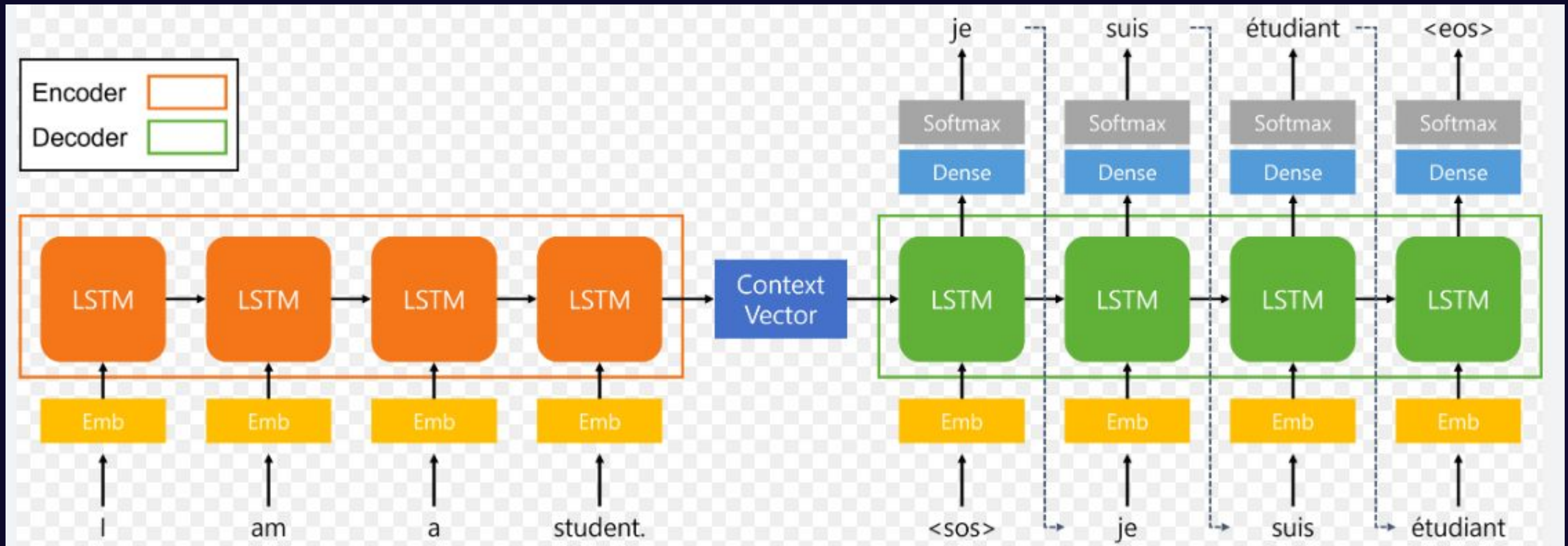


Image Reference: https://medium.com/@vipra_singh/llm-architectures-explained-encoder-decoder-architecture-part-4-b96ace71394c

Reads on Machine Translation and LLMs

1. <https://www.sciencedirect.com/science/article/pii/S2095809921002745>
2. <https://arxiv.org/abs/2004.04902>
3. https://www.researchgate.net/publication/378289524_A_Review_on_Large_Language_Models_Architectures_Applications_Taxonomies_Open_Issues_and_Challenges
4. <https://arxiv.org/pdf/2307.06435>

Recurrent Neural Networks (RNNs)

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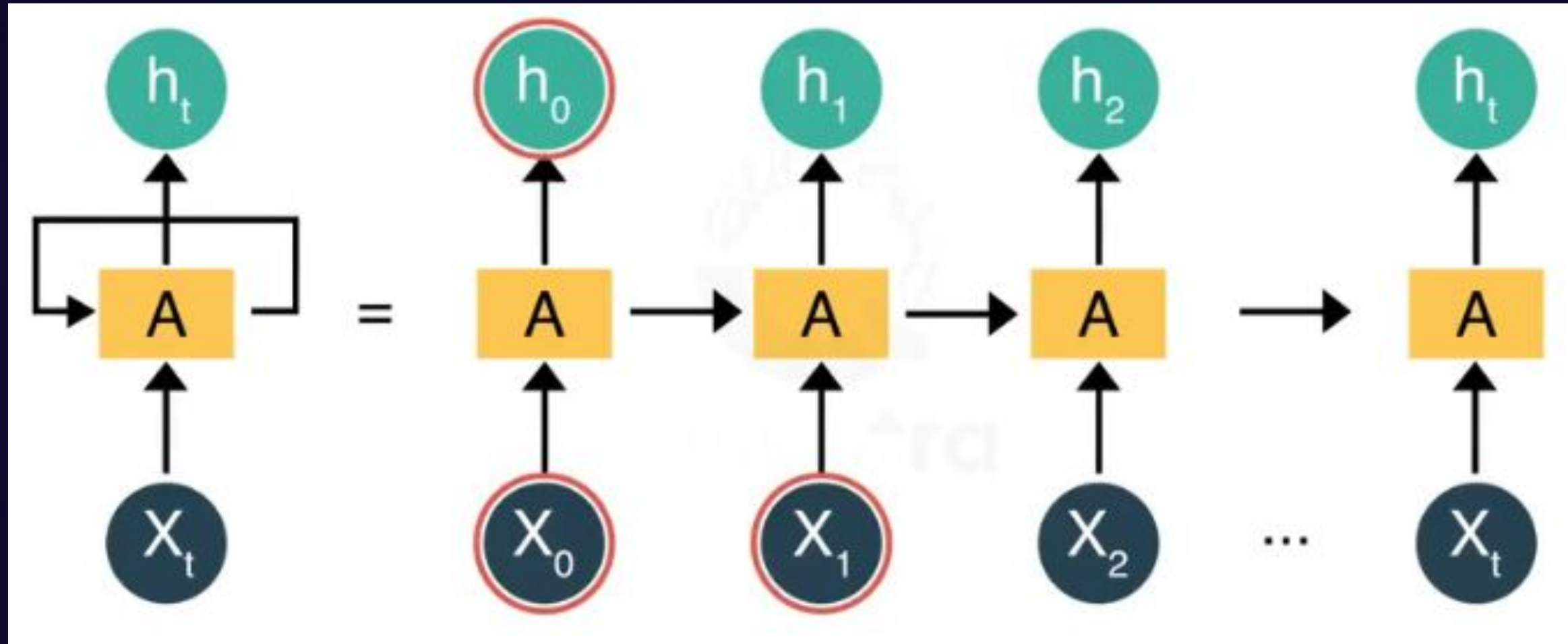


Image Reference: https://quantra.quantinsti.com/startCourseDetails?cid=368§ion_no=4&unit_no=1&course_type=paid&unit_type=WaterMarkVideo

Recurrent Neural Networks (RNNs)

1. Handling sequential data (order is important)
2. Current output is used as an input for the next step
3. Limitation: handling long-term memory
4. Solution: Long-Short Term Memory (LSTM)

Long-Short Term Memory (LSTM)

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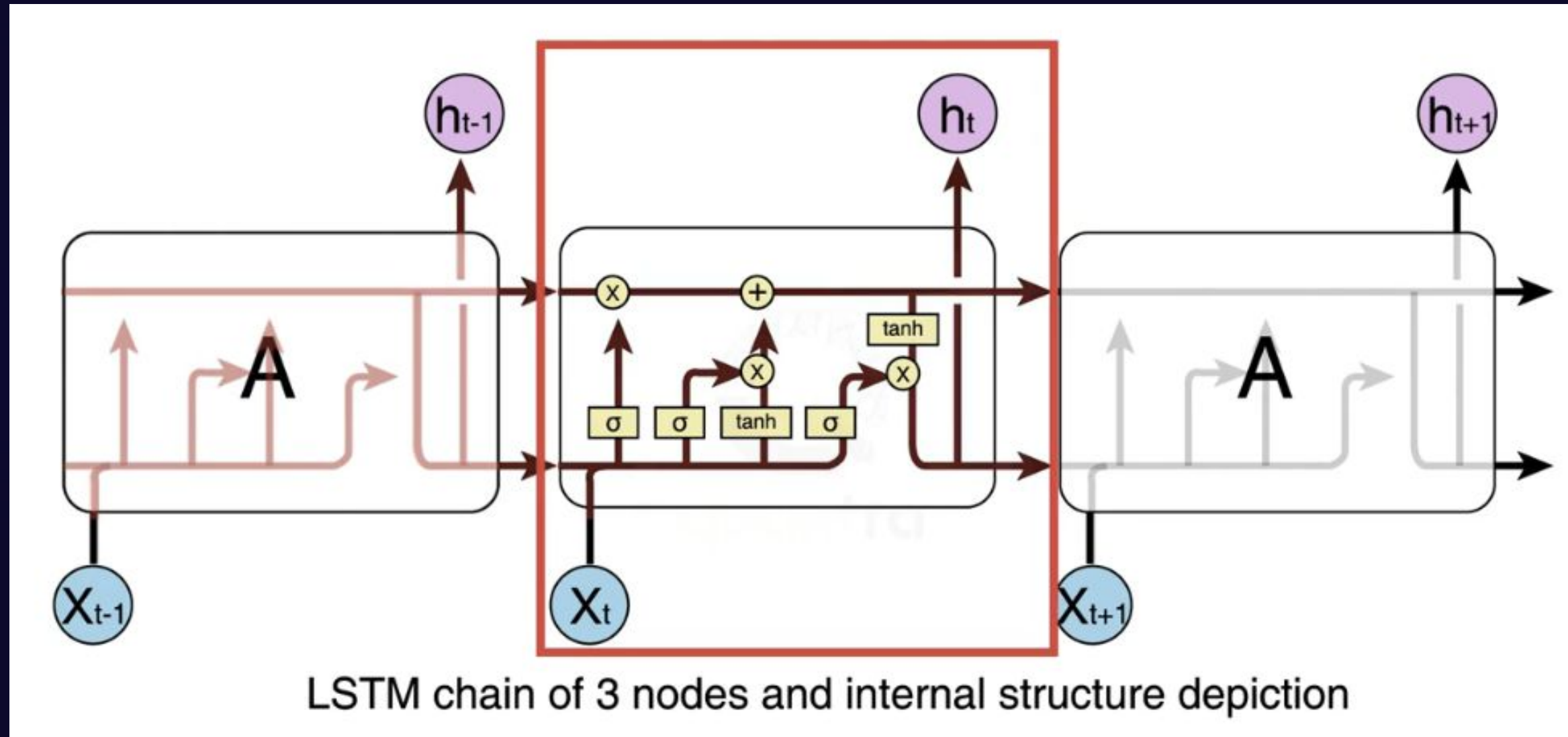


Image Reference: https://quantra.quantinsti.com/startCourseDetails?cid=368§ion_no=4&unit_no=7&course_type=paid&unit_type=WaterMarkVideo

Long-Short Term Memory (LSTM)

1. Gates manage the flow of information
2. Gates decide which information needs to be kept
3. LSTM can still struggle with long sequences, or tasks requiring long memory
4. Solution.....

Attention

Attention

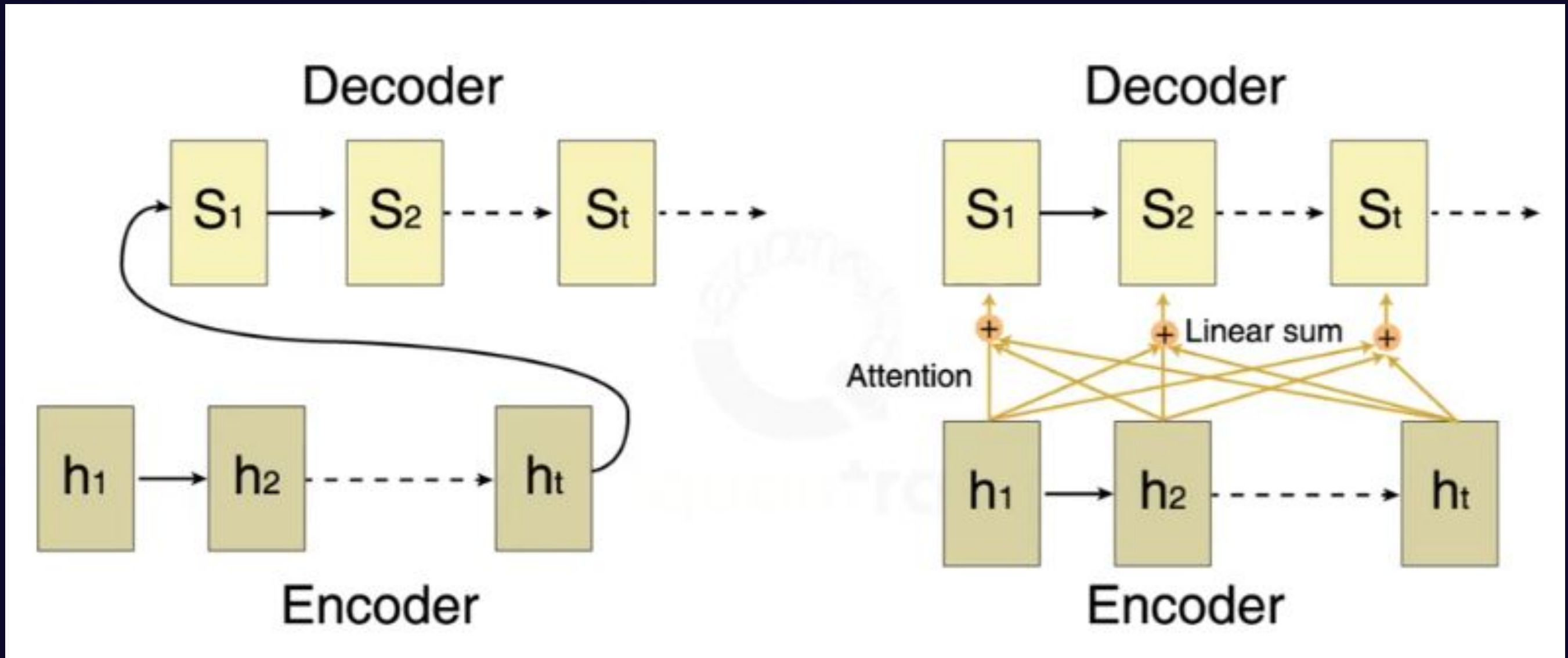


Image Reference: https://quantra.quantinsti.com/startCourseDetails?cid=368§ion_no=5&unit_no=1&course_type=paid&unit_type=WaterMarkVideo

Attention

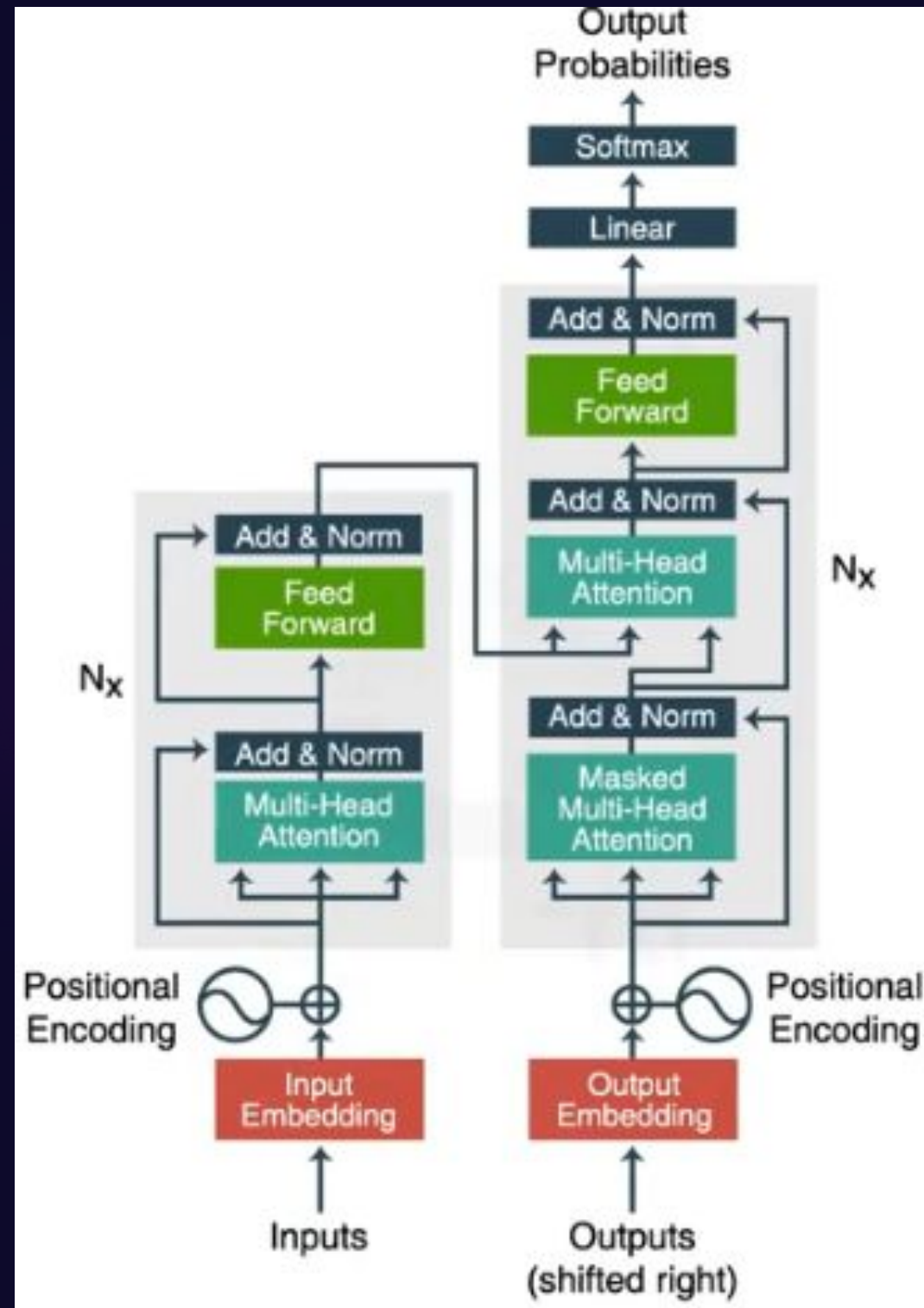
Decoder can access input data

Reads on Attention

1. https://www.researchgate.net/publication/349657026_Attention_Mechanisms_and_Their_Applications_to_Complex_Systems
2. <https://www.mdpi.com/2076-3417/11/9/3883>

Transformer

Transformer



Transformer

1. Only attention
2. Simpler, faster, and cheaper to train
3. Outperformed LSTM+Attention and RNN+Attention
4. Processes input sequences parallelly
5. Positional encoding ensures order information is retained
6. Multi-Head Attention: query, key, and value are inputed to learn relationships between words
7. Add & Norm: normalizes to correct size and shape
8. Nx - repeated N times
9. Masked Multi-Head Attention: prevents look-aheads

Reads on Transformer

1. https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf
2. https://www.researchgate.net/publication/368540509_Transformer_models_an_introduction_and_catalog
3. <https://huggingface.co/docs/transformers/en/index>

Transformers & Attention

Demystifying the Black Box



Sequence Models

How AI processes data over time, evolving from RNNs/LSTMs to modern architectures



Transformer Breakthrough

Processing words in parallel rather than sequentially, enabling massive scale



Attention Mechanism

Understanding context by "looking back" at relevant parts of the prompt

 **Example:** Differentiating "bank" in "river bank" vs. "central bank" by maintaining context across 40-page 10-K filings.

How LLMs Are Built



Pre-training

Reading the internet to learn grammar, facts, and reasoning patterns from billions of documents



Fine-tuning & RLHF

Teaching the model to behave like an assistant using Reinforcement Learning from Human Feedback

Prompt Engineering

Prompt Engineering

Crafting inputs to get specific and useful outputs

Prompt Engineering for Quants

1

Zero-shot vs. Few-shot

Providing examples of input/output to standardize formatting for reliable pipeline integration

2

Chain-of-Thought (CoT)

Forcing step-by-step reasoning to improve logic and accuracy in financial analysis

3

System Prompts

Setting strict personas and constraints: "Output only valid JSON. Do not add conversational filler."

Real Use Cases in Finance

Where Text Becomes Alpha



News Analysis & Sentiment

Extracting specific entity impacts and event classifications beyond simple positive/negative scores



Earnings Call Parsing

Comparing CEO tone across quarters, extracting QA responses on margin pressures instantly



Research Summarization

Synthesizing 50-page sell-side reports into structured bullet points for rapid analysis



Signal Support

Generating alternative data signals as feature engines feeding traditional numerical models

Reads on LLMs in Finance

1. <https://arxiv.org/abs/2303.17564> (BloombergGPT)
2. <https://arxiv.org/abs/1908.10063> (FinBERT)

FinBERT

1. FinBert is fine-tuned on financial data, whereas BERT is a general-purpose model
2. Reuters news articles dataset for pretraining
3. Financial Phrasebank data for fine tuning
4. Used for sentiment analysis
5. <https://medium.com/prosus-ai-tech-blog/finbert-financial-sentiment-analysis-with-bert-b277a3607101>

Live Demo: LLM Trading Pipeline

Downloading the Transcripts:

1. <https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>
2. 'Press Conference'
3. 'Transcript (PDF)'

Python:

1. Set up the environment
2. Create an account with Alpaca brokers for the API keys (optional, but recommended)
3. Data Collection and Processing.ipynb
4. Sentiment Score of FOMC Transcripts.ipynb
5. Trading Strategy Based on Sentiment Score Threshold.ipynb



Production Challenges

The Harsh Reality of Trading Environments

Hallucinations

LLMs sound confident even when fabricating financial metrics or events

Latency

Generation takes seconds; market opportunities disappear in milliseconds

Model Risk & Drift

Behavior changes silently when providers update underlying models

Compliance & Privacy

Sending proprietary strategies to external APIs introduces infosec risks

Cost

Processing millions of articles daily at scale is financially intensive

Key Takeaways

- Build mental models

Understand transformers and attention with minimal theory
- Know where LLMs excel

Research tasks like news analysis, earnings parsing, and summarization
- Master prompt design

Reduce noise and hallucinations for finance use cases
- Recognize constraints

Latency, compliance, and production limitations in trading environments

